RESCUELOGIC®

The SAFER Protocol

Security And Fire Emergency Reports

Payload Structure for Life-Safety Event Meaning

CORE POSITIONING SAFER standardizes life-safety event meaning, not life-safety systems. Transport moves the message. SAFER defines what the message means.

READER-FRIENDLY STATEMENT The pattern was already there. SAFER makes it visible.

Version 1.1 • Proposed Specification

October 2026

Original protocol © 2024 Dan Horon; v1.1 revision October 2026
Licensed under the Apache License, Version 2.0
U.S. Copyright Registration No. TXu 2-426-289
Effective March 7, 2024

Contents

Section	Topic
1	Introduction
2	Version 1.1 Summary
3	Scope: Payload Only
4	Core Event Information Model
5	Required SAFER Fields
6	STATUS Values
7	Optional Fields and Extensibility
8	Example SAFER Payloads
9	Legacy Formats and Contact ID
10	Transport, Security, and Reliability
11	Implementation and Testing Guidance
12	Baseline Conformance Profile
13	Governance, Stewardship, and Licensing
Appendix A	Minimal JSON Schema for Testing
Appendix B	Field Reference Summary
Appendix C	Reference Materials and Contact Information

1. Introduction

The Security And Fire Emergency Reports (SAFER) Protocol defines a structured JSON payload for reporting life-safety and security events.

A payload is the content of a message. SAFER names the key parts of a reported event so different systems and people can preserve the same meaning as the information moves between software, monitoring platforms, building records, and response tools.

PLAIN-LANGUAGE POSITION SAFER standardizes life-safety event meaning, not life-safety systems. Transport moves the message. SAFER defines what the message means.

SAFER is designed for modern networked systems, but it does not choose the network. It does not define a fire alarm system, monitoring-station connection, communication pathway, listed product, or code-compliant installation. It is a way to represent a reported event clearly and consistently.

CORE EVENT SENTENCE This system reports that this device, of this type, at this location, has changed to this state.

1.1 The pattern was already there

Fire alarm panels from different manufacturers use different words, formats, and message styles. When their outputs are compared side by side, their reported events repeatedly answer the same basic questions: who reported, which device, what kind of device, where it is, and what changed.

SAFER does not force manufacturers to become identical. It gives the meaning they already report a common structure that can be tested, validated, shared, and improved.

2. Version 1.1 Summary

Version 1.1 clarifies the proposed SAFER event-payload specification. It retains the seven required fields and the six numeric STATUS meanings of v1.0 while aligning terminology with SAFER Common Vocabulary. It does not adopt a separated condition/qualifier wire format.

2.1 Compatibility intent

Version 1.1 retains the baseline required-field shape and STATUS codes of v1.0. A payload must still declare its actual SAFER_ID.version. This document makes no assertion about the number of external deployments of earlier versions.

2.2 Main clarifications

- SAFER standardizes event meaning and payload structure, not fire alarm systems themselves.
- Transport and meaning are separate. MQTT, HTTP, AMQP, TCP/IP, serial gateways, files, or other methods may carry the same SAFER payload.
- Legacy formats such as Contact ID can coexist with SAFER and may be wrapped, translated, or preserved as optional reference data.
- SAFER does not define pathway supervision, transport reliability, equipment listing, NFPA compliance, UL evaluation, supervising-station requirements, or emergency response policy.
- A change of condition is the conceptual event model. A transmitted event report may instead describe a restoration, repetition, or status observation. Such reports should not be assumed to represent a new physical activation.
- Developers should use standard JSON parsers and schema validation for testing.

VERSIONING NOTE Version 1.1 introduces clarifications and backward-compatible additions to Version 1.0. Future backward-compatible changes may use a minor version increment. Changes that break existing required-field compatibility require a new major version, such as Version 2.0.

3. Scope: Payload Only

SAFER defines the structure of a reported event payload. In this document, a SAFER payload is a JSON object containing defined fields that preserve the meaning of a life-safety or security event.

SAFER defines	SAFER does not define
Required and optional JSON fields for event representation.	Network architecture or communication pathway design.
Common field meanings for reporting context, device identity, device type, device location, and event status.	Transport protocols such as MQTT, HTTP, AMQP, TCP, UDP, serial, cellular, or IP dialer-capture methods.
Message versioning and unique event identification through SAFER_ID.	Pathway supervision, survivability, redundancy, or network performance requirements.
An extensible structure that can add context without breaking the baseline payload.	Product listing, equipment certification, UL/ULC evaluation, or code compliance.
A mapping target for legacy or manufacturer-specific event messages.	Supervising-station requirements, emergency response procedures, or manufacturer internal architecture.

PAYLOAD-ONLY PRINCIPLE SAFER can be tested, validated, and adopted as an event payload without changing listed equipment, pathway supervision methods, product architecture, or compliance responsibilities.

4. Core Event Information Model

The SAFER event model begins with a practical observation: alarm systems already contain a small set of recurring information roles. Different panels may display or transmit those facts differently, but the same roles keep appearing.

$$e_t = (\text{System}, \text{Identity}, \text{DeviceType}, \text{Location}, \text{Condition}_t)$$

The notation is useful for technical review, but readers do not need to use it to understand SAFER. In plain language, the model says:

PRIMITIVE EVENT STATEMENT This system reports that this device, of this type, at this location, has changed to this state.

4.1 Stable object and changing state

The recurring five-role model describes the identity and condition of a known point. Not every message announces a new change: a report can describe activation, restoration, a repeated condition, or a status-only observation.

$$\text{DEVICE} = (\text{ID}, \text{TYPE}, \text{LOCATION})$$
$$\text{REPORT CODE} = \text{SAFER STATUS}$$
$$\text{EVENT} = \Delta\text{STATE} \times \text{DEVICE} \times \text{TIME}$$

DEVICE_ID identifies a point within its reporting/source context; TYPE and LOCATION describe that point and may be corrected over time. The conceptual expression EVENT = ΔSTATE × DEVICE × TIME describes a change, not arithmetic. In v1.1, SAFER STATUS is a six-value reporting code encompassing condition/classification and reporting relationship; it is not always a persistent device state. Optional fields may preserve source details, times, and descriptive condition text.

4.2 Classification and descriptive condition

A reported condition can serve more than one information need. A normalized reporting code gives software a consistent value to interpret. A descriptive condition supplies more specific information from the source, which may explain why the condition was reported.

Receiving applications may map reporting codes into a smaller set of human-facing states while preserving specific condition text. For example, RescueLogic uses three Base States—Normal, Alarm, and Trouble—to select recognizable device-state images. Separately extracted condition text helps explain what the source reported.

SAFER does not require recipients to use RescueLogic's three-state model or any particular display method. Its purpose is to preserve information that different applications can interpret and present appropriately.

4.3 Mapping the model to SAFER fields

Information role	SAFER field	Question	Example
Reporting context	CUSTOMER + PANEL_ID	Who is reporting?	Account 1234 / Node 01
Identity	DEVICE_ID	Which device or point?	L1S067
Device Type	DEVICE_TYPE	What kind of thing?	Smoke Detector
Location	DEVICE_LOCATION	Where does it belong?	Classroom 4 West
Reporting code	STATUS	How is this report classified in SAFER?	4 = Trouble

The optional CONDITION field can carry more specific descriptive information about what the source reported. STATUS supplies the defined numeric reporting meaning; CONDITION adds detail without changing that meaning.

5. Required SAFER Fields

A minimum SAFER Version 1.1 payload contains the fields below. Required means required for a baseline SAFER payload. It does not mean a code requirement, product listing requirement, or regulatory mandate.

Field	Type	What it means	Example	Plain-language note
CUSTOMER	string	Customer, account, facility, or reporting relationship.	1234	Sets the overall reporting context.
PANEL_ID	string	Panel, node, system, or source device.	Node 01	Tells which source reported the event.
DEVICE_ID	string	Device, point, zone, module, address, or logical point.	L1S067	Preserve the original device identity when possible.
DEVICE_TYPE	string	Kind of device or point.	Smoke Detector	Device type gives meaning that STATUS alone cannot carry.
DEVICE_LOCATION	string	Location text for the device or point.	Classroom 4 West	Often the site-programmed device location text.
STATUS	integer	Normalized SAFER event status code.	1	Defined values are listed in Section 6.
SAFER_ID	object	SAFER version and unique identifier for the report record.	{"version":"1.1", "..."}	Supports review, tracing, and compatibility.

FIELD NAMING CONVENTION SAFER field names use uppercase so defined life-safety event fields are easy to recognize. Uppercase is only a naming convention; the field meaning comes from the SAFER definition and schema.

6. STATUS Values

STATUS is the normalized SAFER reporting code. Its six v1.1 meanings intentionally retain the existing mixture of active conditions, supervisory/trouble classifications, restoration, status-only reports, and previously reported conditions. This is a compatibility-oriented vocabulary, not a claim that these are mutually exclusive dimensions of physical state.

A human-facing display may use its own classifications, words, colors, or icons. Do not equate SAFER STATUS numbers with Contact ID qualifiers or RescueLogic Base State and LogicalState keys. The question of separating reported condition from event qualifier is open for a subsequent version.

Code	SAFER meaning	Typical use
1	Active	A device or point is reporting its active condition. The receiving system may classify it as fire alarm, security, medical, or another category.
2	Supervisory	A supervisory condition is being reported.
3	Restore	A prior condition is returning toward normal. Restore is an event status, not a replacement for event history.
4	Trouble	A trouble condition is reported for the system, a point, a pathway, or a monitored condition. This includes cases where expected operation cannot be confirmed.
5	Status only	A factual status message. It should not automatically be treated as a new alarm, supervisory, or trouble event.
6	Previously reported	The condition was previously reported and is being repeated or referenced.

MACHINE VALUE A query such as STATUS = 1 has one defined SAFER meaning. A user interface may display ACTIVE, ALARM, local wording, or site-specific labels without changing the machine value.

6.1 From reporting code to human presentation

SAFER STATUS identifies the normalized meaning of a report. A receiving application may interpret that report in relation to a known device type, its previous reported condition, and available source information.

RescueLogic provides one implementation example. Its Script Panel Types configuration uses the Event Type to identify a Base State of Normal, Alarm, or Trouble. Device State Images can then present that state using an image associated with the device type. Separately, it extracts a descriptive Condition text if available to help the operator understand the reported situation.

This demonstrates an important distinction: a small number of recognizable state categories can support a much larger variety of descriptive conditions. The state helps people recognize the situation; the condition text helps explain what the source reported.

SAFER STATUS values are not numerically equivalent to RescueLogic Base State keys. In particular, a SAFER Restore, Status only, or Previously reported message may require event history or source context before a receiving application can determine the appropriate displayed state.

7. Optional Fields and Extensibility

SAFER uses an OPTIONAL object to add context, preserve source details, support migration, and improve presentation. OPTIONAL fields must not change the defined meaning of required fields. When adding information from a point list or other lookup, distinguish what the source reported from what was referenced or derived.

Systems should ignore optional fields they do not understand, while preserving them where practical for logs, review, and forwarding.

Optional field	Type	Use
PANEL_GPS	object	GPS position of the panel or source, if useful. Example: {"latitude":"37.7749","longitude":"-122.4194"}
LOCATION	string	General area, floor, building, or level. Example: 2nd Floor
CONDITION	string	Human-readable description of the specific condition or event detail reported by the source, or an explicitly identified interpretation of that information. May provide a reason or useful clue about the STATUS, particularly for trouble reports. Examples: Open Circuit, Ground Fault, Detector Missing. Does not override the meaning of STATUS.
OWNER_LOCATION	string	Owner- or responder-friendly location. Example: Conference Room
EVENT_TIME	string	Timestamp when the event was observed or reported. Prefer ISO 8601. Example: 2026-10-07T10:14:00-04:00
DATE	string	Optional separate date field for older systems. Example: 2026-10-07
TIME	string	Optional separate time field for older systems. Example: 10:14:00
SYSTEM_CATEGORY	string	General system category. Example: Fire Alarm System
ZONE	object	Zone context, name, notes, or URL. Example: {"name":"Zone B2"}
GROUP	object	Building, campus, tenant, or grouped area. Example: {"name":"North Campus"}
GPS_COORDINATES	object	GPS position of the event or device, when useful.
CONTACT_ID	object	Legacy Contact ID reference. Example: {"version":"SIA DC-05","code":"1234181101010158"}
SOURCE_FORMAT	string	Name of the source format or native message family. Example: Panel printer text
SOURCE_MESSAGE	string	Original source message or raw event text for audit and mapping. Example: N01-L1:001 ALARM ON PHOTO

EXTENSIBILITY RULE Do not use optional fields to change the meaning of required fields. Optional fields should add context, preserve source information, or support a specific product, project, or software system.

8. Example SAFER Payloads

These examples are illustrative. Implementers should validate actual payloads against the adopted schema and project requirements.

8.1 Minimal smoke detector active event


```
{
  "CUSTOMER": "1234",
  "PANEL_ID": "Node 01",
  "DEVICE_ID": "L1S067",
  "DEVICE_TYPE": "Smoke Detector",
  "DEVICE_LOCATION": "Classroom 4 West",
  "STATUS": 1,
  "SAFER_ID": {
    "version": "1.1",
    "unique_id": "123e4567-e89b-12d3-a456-426614174000"
  }
}
```


8.2 Comparable active report with optional context


```
{
  "CUSTOMER": "1234",
  "PANEL_ID": "Node 01",
  "DEVICE_ID": "L1S067",
  "DEVICE_TYPE": "Smoke Detector",
  "DEVICE_LOCATION": "Classroom 4 West",
  "STATUS": 1,
  "SAFER_ID": {
    "version": "1.1",
    "unique_id": "223e4567-e89b-12d3-a456-426614174000"
  },
  "OPTIONAL": {
    "SYSTEM_CATEGORY": "Fire Alarm System",
    "CONDITION": "Smoke Detected",
    "LOCATION": "First Floor",
    "OWNER_LOCATION": "Classroom 4 West",
    "EVENT_TIME": "2026-10-07T10:14:00-04:00",
    "SOURCE_FORMAT": "Panel printer text",
    "SOURCE_MESSAGE": "N01-L1:067 ALARM ON PHOTO CLASSROOM 4 WEST"
  }
}
```

8.3 Valve supervisory event


```
{
  "CUSTOMER": "1234",
  "PANEL_ID": "Node 01",
  "DEVICE_ID": "L1M082",
  "DEVICE_TYPE": "Valve Supervisory Switch",
  "DEVICE_LOCATION": "Riser Room",
  "STATUS": 2,
  "SAFER_ID": {
    "version": "1.1",
    "unique_id": "323e4567-e89b-12d3-a456-426614174000"
  }
}
```


8.4 Trouble Report with Descriptive Condition


```
{
  "CUSTOMER": "1234",
  "PANEL_ID": "Node 01",
  "DEVICE_ID": "L1M083",
  "DEVICE_TYPE": "Monitor Module",
  "DEVICE_LOCATION": "Riser Room",
  "STATUS": 4,
  "SAFER_ID": {
    "version": "1.1",
    "unique_id": "423e4567-e89b-42d3-a456-426614174000"
  },
  "OPTIONAL": {
    "CONDITION": "Open Circuit",
    "SOURCE_FORMAT": "Panel printer port text (RS232)",
    "SOURCE_MESSAGE": "NODE 01 L1M083 TROUBLE OPEN CIRCUIT"
  }
}
```


In this illustrative message, STATUS=4 identifies the report as Trouble. CONDITION="Open Circuit" preserves a more specific description of the reported trouble.

A receiving application can present the monitor module in its Trouble state while displaying Open Circuit to help the user understand the source's report.

Another trouble report may use the same STATUS value with a completely different CONDITION description. The device's type and identity remain separately available.

Neither the STATUS code nor the descriptive condition alone proves the physical cause of the reported trouble.

9. Legacy Formats and Contact ID

SAFER is not a mandate to eliminate existing alarm communication formats. Legacy formats such as Contact ID remain widely understood and may continue to be used where appropriate.

SAFER provides a structured event payload that can coexist with, wrap, translate, or reference legacy formats.

COEXIST • WRAP • TRANSLATE • MIGRATE SAFER provides a migration path to richer structured reporting without requiring installed systems or monitoring workflows to change all at once.

Compact legacy reports may require receiver lookup tables, source configuration, or point records for interpretation. SAFER can carry reporting context, device identity, type, location, and STATUS together, but a translator must not invent a device type or location when the available evidence cannot establish one.

Legacy part	Common limitation	SAFER representation
Account or subscriber ID	May identify the account but not the panel or node.	CUSTOMER + PANEL_ID
Zone/contact/point value	May need external mapping to recover the original point identity.	DEVICE_ID
Event code	May require code tables and receiver interpretation.	Interpret source event code and Q qualifier with source context; STATUS plus optional CONDITION and CONTACT_ID
Location text	Often not carried in a compact report.	DEVICE_LOCATION and optional OWNER_LOCATION
Original code	May still matter for compatibility and review.	OPTIONAL.CONTACT_ID

In Contact ID, Q=1 means a new event/opening, Q=3 a new restore/closing, and Q=6 a previously reported condition still present. These are Contact ID qualifier meanings, NOT SAFER STATUS mappings. Interpret the qualifier alongside the event code and partition-zone/user context. Preserve the original report and identify referenced or derived enrichment where practical; unclear or missing point information should remain subject to review.

9.1 Contact ID translation guidance

Contact ID translation is source-aware, not a direct substitution of Q qualifiers for SAFER STATUS values. Mapping rules should be documented and tested against source reports and known configuration.

Interpret event codes, Q qualifiers, and partition/zone or user context together. Preserve the original report and identify any point records or configuration used to supply device type or location.

Verified mapping examples and adapter tests may be published as repository companions. For v1.1, the required fields, STATUS definitions, and baseline conformance rules remain in Sections 5, 6, and 12.

10. Transport, Security, and Reliability

SAFER is independent of the transport method. A SAFER payload may be carried over MQTT, HTTP, AMQP, TCP, UDP, serial gateways, files, message queues, APIs, or other transport mechanisms.

SAFER does not make a transport reliable, supervised, secure, listed, or code-compliant by itself.

> **TRANSPORT ≠ MEANING** A transport can deliver a message successfully while leaving application meaning undefined. SAFER supplies the event meaning; the surrounding system supplies transport, supervision, security, and compliance mechanisms.

System designers, manufacturers, integrators, and owners remain responsible for applicable requirements involving pathway supervision, survivability, encryption, authentication, access control, network segmentation, monitoring-station operation, product listing, and approval by authorities having jurisdiction.

10.1 Security considerations

- Protect SAFER payloads in transit and at rest according to the risk and application.
- Authenticate sources and receivers where SAFER payloads influence operations or records.
- Validate input data before display, storage, forwarding, or automated processing.
- Log source information and transformation steps where SAFER payloads are mapped from another format.
- Do not treat SAFER publication as approval of any specific security architecture.

11. Implementation and Testing Guidance

Developers should implement SAFER using ordinary JSON tooling. The payload is intentionally simple so it can be generated, parsed, validated, logged, transformed, and tested with widely available tools.

11.1 Recommended implementation steps

1. Generate a minimal SAFER event using the required fields.
2. Validate that the JSON is syntactically correct and that required fields are present.
3. Validate STATUS against the defined values.
4. **Preserve the native source report, its qualifier and event code where available, and the provenance of any lookup or derived values.**
5. Add optional fields only when they improve context, interoperability, review, or presentation.
6. Test payloads with more than one parser, programming language, or receiving tool when interoperability matters.

11.2 Parsing guidance

Production software should use JSON parsers rather than regular expressions for SAFER messages. Regular expressions can be useful for log review or legacy text extraction, but they should not be the primary method for validating structured JSON payloads.

11.3 Validation guidance

- Reject malformed JSON, or hold it for review.
- Reject baseline SAFER payloads that are missing required fields.
- Reject STATUS values outside the defined range.
- Warn when optional fields conflict with required fields or source information.
- Preserve uncertainty rather than forcing incomplete or conflicting information into false certainty. Required fields establish structural conformance, not verification that the associated physical device description is correct.

TRUTH-PRESERVING VALIDATION A system should not invent certainty. When data is missing or conflicting, the honest result may be warning, needs review, or MAYBE. Fire alarm systems already teach this lesson: they report trouble when the system cannot fully trust its own ability to operate.

CONDITION DESCRIPTIONS When reports have the same STATUS but different CONDITION descriptions, a receiving application can preserve and distinguish those descriptions. Even when both reports use the same broad state or device image, the descriptive condition can remain available to technicians, operators, and other authorized recipients.

This is recommended implementation guidance, not an additional baseline conformance requirement.

12. Baseline Conformance Profile

In this section, conformance means that a payload follows the SAFER Version 1.1 format. It does not indicate that a fire alarm system, security system, network pathway, supervising-station connection, listed product, or emergency response procedure is compliant with any code or listing standard.

A baseline SAFER Version 1.1 payload is conforming when all of the following are true:

- The payload is valid JSON.
- The payload contains CUSTOMER, PANEL_ID, DEVICE_ID, DEVICE_TYPE, DEVICE_LOCATION, STATUS, and SAFER_ID.
- SAFER_ID contains version and unique_id. For baseline v1.1 conformance, version shall be the string "1.1", and unique_id shall be a nonempty string. The issuer is responsible for ensuring report-record identifier uniqueness.
- For baseline SAFER Version 1.1 conformance, STATUS shall be one of the six defined integer values: 1, 2, 3, 4, 5, or 6. Implementations may define additional behavior through documented extensions, but an extension does not silently change the baseline meanings or valid STATUS values.
- Optional fields do not change the meaning of required fields.
- Optional fields, if present, are structured consistently with this specification or an identified extension.

12.1 Recommended test cases

Test case	Expected result	Purpose
Minimal valid event	Accepted	Baseline compatibility
Missing DEVICE_LOCATION	Rejected or marked invalid	Required-field validation
STATUS = 9	Invalid for baseline SAFER v1.1	Status validation
SAFER_ID.version = "0.0"	Invalid for SAFER v1.1	Version validation
SAFER_ID.unique_id = ""	Invalid for baseline SAFER v1.1	Nonempty report identity
Unknown optional field	Preserved or ignored without failing the baseline event	Extensibility
Malformed JSON	Rejected or held for review	Syntax safety
CONTACT_ID inside OPTIONAL	Accepted	Migration support
SOURCE_MESSAGE included	Accepted and preserved where practical	Review and troubleshooting

12.2 Open design question: condition and qualifier

This is a documented design question, not an adopted v1.1 wire-format change. The present STATUS field can represent “Supervisory” or “Previously reported,” but a report can truthfully have both properties. A future representation may separate the reported condition or classification from the qualifier that says whether the report is new, restored, repeated, or status-only.

Two concerns are legitimate: (A) keep a simple six-value code to protect ease of adoption and existing implementations; (B) preserve independent dimensions so translation and future sensors do not lose meaning. Possible paths include retaining the combined codes, adding an optional qualifier, or defining separate fields in a versioned migration.

Examples showing the relationship between a reported condition and an event qualifier are welcome through GitHub Issues. Useful examples include new, restored, repeated, and status-only reports. Please remove customer-identifying information and include the expected interpretation and any compatibility concerns. Proposed changes can be discussed and tested before they are considered for a future version. Any change to required meanings or wire format must follow the versioning rules in Section 13.

13. Governance, Stewardship, and Licensing

The SAFER Protocol is published as an open specification under the Apache License 2.0. The intent is to allow manufacturers, monitoring platforms, system integrators, software developers, researchers, and standards participants to review, test, implement, and improve the structure without licensing barriers.

13.1 Versioning

Version type	Use when	Compatibility expectation
Patch	Typographical corrections or non-substantive clarifications.	No payload compatibility impact.
Minor	Backward-compatible clarifications, optional fields, test profiles, or guidance.	Existing required-field payloads remain valid.
Major	Required fields, field meanings, or status values change in a way that breaks existing implementations.	Migration guidance required.

13.2 Stewardship

Future revisions may incorporate contributed examples and tests of the distinction between reported condition/classification and event qualifier. SAFER Common Vocabulary is an explanatory companion and discussion reference; it does not silently amend the v1.1 baseline payload.

Governance may evolve through an open working group, collaborative industry forum, standards development engagement, or another transparent process.

13.3 No endorsement implied

Publication of this protocol does not imply endorsement, approval, certification, or listing by NFPA, UL, ULC, any standards development organization, authority having jurisdiction, manufacturer, monitoring company, or regulatory body.

Appendix A — Minimal JSON Schema for Testing

The following minimal schema is for structural testing. It checks required fields, version, and basic reporting codes while allowing additional properties for extensions. The full specification governs field meaning. Schema validation does not verify physical conditions, source interpretation, or unique_id uniqueness across reports.

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "SAFER Event Payload v1.1",
  "type": "object",
  "required": [
    "CUSTOMER",
    "PANEL_ID",
    "DEVICE_ID",
    "DEVICE_TYPE",
    "DEVICE_LOCATION",
    "STATUS",
    "SAFER_ID"
  ],
  "properties": {
    "CUSTOMER": {
      "type": "string"
    },
    "PANEL_ID": {
      "type": "string"
    },
    "DEVICE_ID": {
      "type": "string"
    },
    "DEVICE_TYPE": {
      "type": "string"
    },
    "DEVICE_LOCATION": {
      "type": "string"
    },
    "STATUS": {
      "type": "integer",
      "enum": [
        1,
        2,
        3,
        4,
        5,
        6
      ]
    },
    "SAFER_ID": {
      "type": "object",
      "required": [
        "version",
        "unique_id"
      ],
      "properties": {
        "version": {
          "type": "string",
          "const": "1.1"
        },
        "unique_id": {
          "type": "string",
          "minLength": 1
        }
      }
    },
    "OPTIONAL": {
      "type": "object"
    }
  },
  "additionalProperties": true
}
```

Appendix B — Field Reference Summary

Field	Required?	Meaning	Notes
CUSTOMER	Yes	Customer, account, facility, or reporting relationship.	Defines overall context.
PANEL_ID	Yes	Panel, node, system, or source identifier.	Part of reporting context.
DEVICE_ID	Yes	Device, point, zone, address, module, or logical point identifier.	Preserve source identity when possible.
DEVICE_TYPE	Yes	Kind of device or point reporting.	Helps interpret the event.
DEVICE_LOCATION	Yes	Location text associated with the device or point.	Often site-programmed text.
STATUS	Yes	Normalized SAFER event status code.	1 Active, 2 Supervisory, 3 Restore, 4 Trouble, 5 Status only, 6 Previously reported.
SAFER_ID.version	Yes	SAFER message version.	Use 1.1 for this version.
SAFER_ID.unique_id	Yes	Unique identifier associated with the SAFER report record.	Supports traceability.
OPTIONAL	No	Container for additional context.	Do not override required meaning.
OPTIONAL.CONDITION	No	Specific human-readable condition detail	Adds information without overriding STATUS
OPTIONAL.CONTACT_ID	No	Legacy Contact ID reference.	Supports coexistence, migration, and review.
OPTIONAL.SOURCE_MESSAGE	No	Original native message text.	Supports troubleshooting and mapping validation.

Appendix C — Reference Materials and Contact Information

Reference materials

- SAFER Protocol v1.0, Security And Fire Emergency Reports, published under the Apache License 2.0.
- SAFER Protocol FAQ.
- SIA Contact ID Protocol for Alarm System Communications.
- NFPA 72, National Fire Alarm and Signaling Code. Project requirements should be verified against the edition adopted by the Authority Having Jurisdiction.
- NFPA 950, Standard for Data Development and Exchange for the Fire Service.
- RescueLogic publications and implementation materials.

Contact and stewardship

Questions, observations, and proposed improvements are welcome through GitHub Issues and Pull Requests. Direct inquiries may also be sent to:

Contact	Email
Dan Horon	dan@rescuelogic.com
Jeff Horon	jeff@rescuelogic.com

https://github.com/rescuelogic/SAFER